# Welcome to
# **Create Custom Map with *QGIS* Print Layout**

Questions to think about:

- What is your name?
- What program/department are you from?
- What brought you to the workshop today?
- What do you hope to get out the workshop?

2026

University of Victoria Libraries

With *QGIS* someone could:

- Create beautiful maps…

And/or

- Conduct spatial analysis…

QGIS has hundreds of tools and possibilities…

**This is an Introduction to *QGIS* with creating a custom map**

# Create Custom Map with *QGIS* Print Layout

2026

University of Victoria Libraries

As per the instructions when you signed up:

**QGIS downloaded on your computer?**

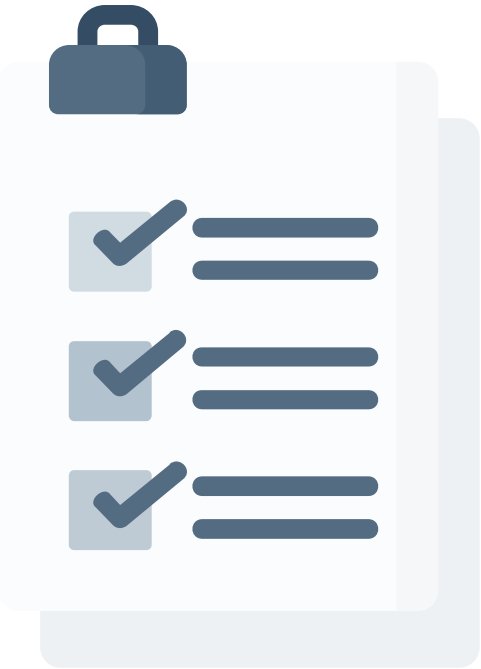
If not, please come back to the workshop another time!

# Territorial Acknowledgment

**We acknowledge and respect the lək̓ʷəŋən peoples on whose traditional territory the university stands and the Songhees, Esquimalt, and WSÁNEĆ peoples whose historical relationships with the land continue to this day.**

# Learning Objectives

- **Review** and list differences with some mapping tools
- **Identify** and **navigate** key *Print Layout* elements
- **Distinguish** why *QGIS Print Layout* is necessary for custom map(s)



# Outcomes

Using *QGIS*, participants will:

- **Load** (.shp, .csv) files
- **Navigate** main features of *Print Layout*
- **Create** and **export** custom map using *QGIS Print Layout*

# What is a Mapping Tool?

- Software/hardware typically designed for specific tasks or functions such as mapping
- Very limited spatial analysis

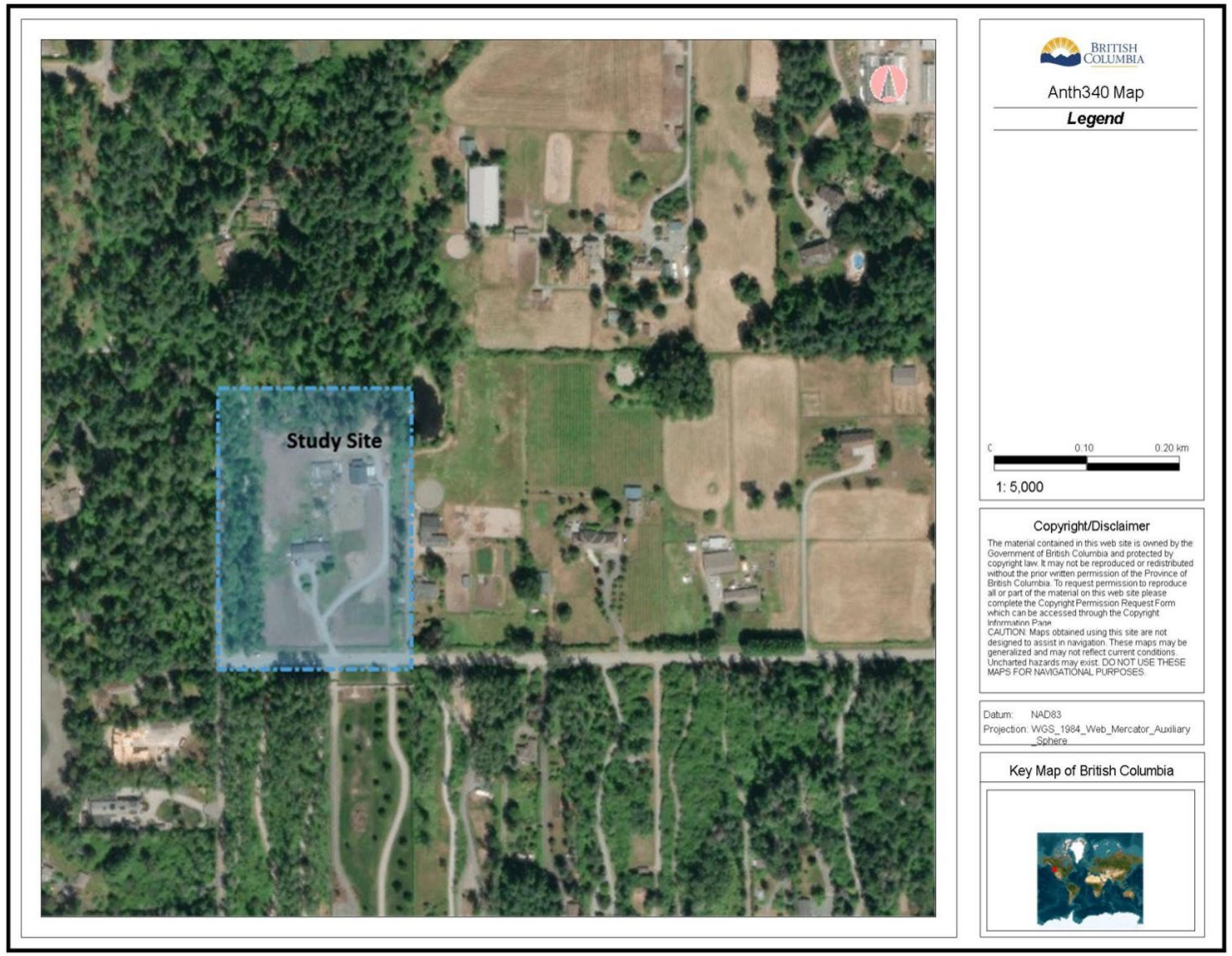
# What is a GIS?

- A comprehensive system which encompasses a range of geospatial tools for spatial modelling and analysis

**Not all mapping tools are a GIS!**

# Maps from mapping tools

Some applications can make very basic maps, but lack tools to create customizable, advanced maps like *QGIS*




Map created on *iMapBC*

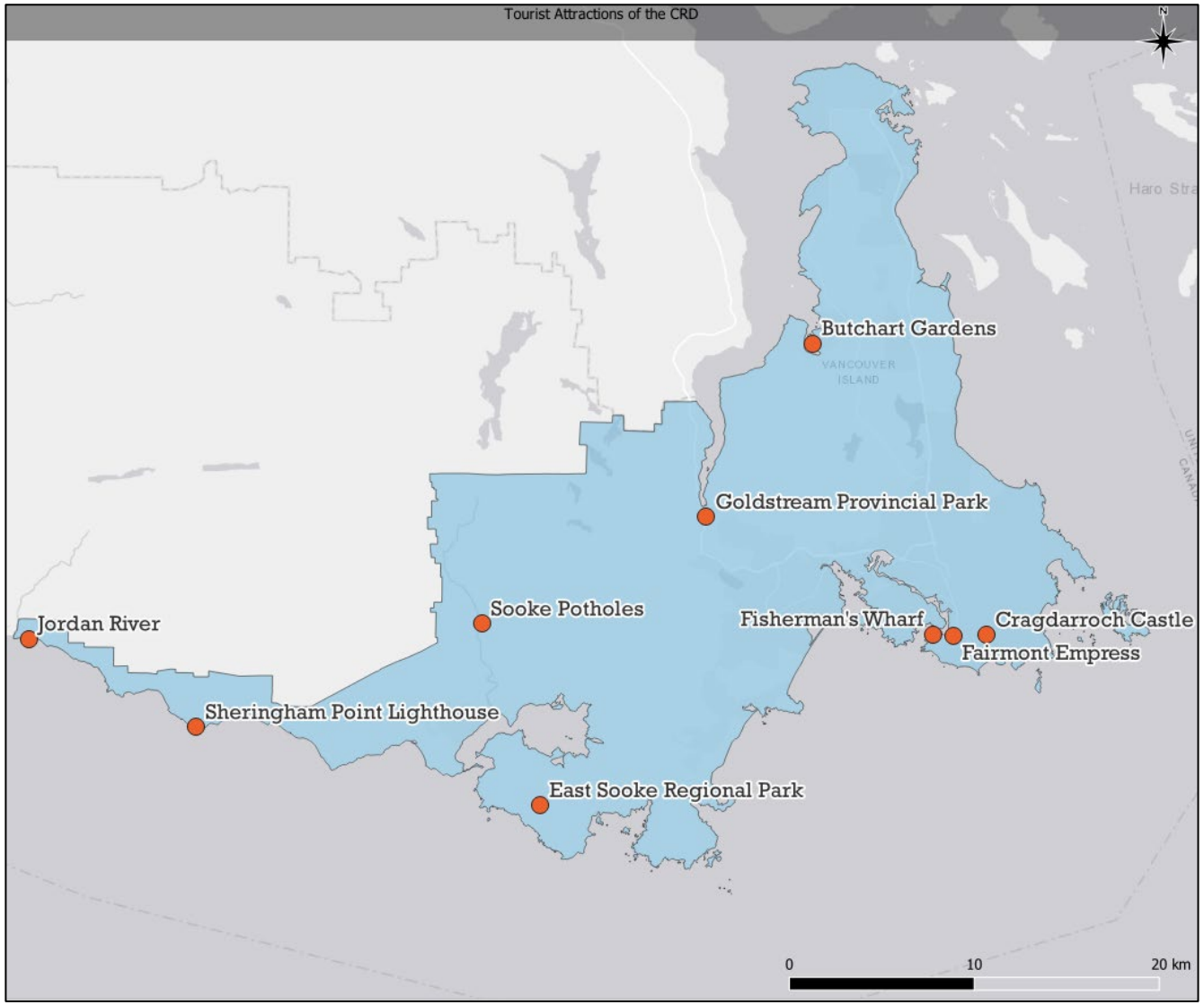
# Maps from mapping tools

Some applications can make very basic maps, but lack tools to create customizable, advanced maps like *QGIS*

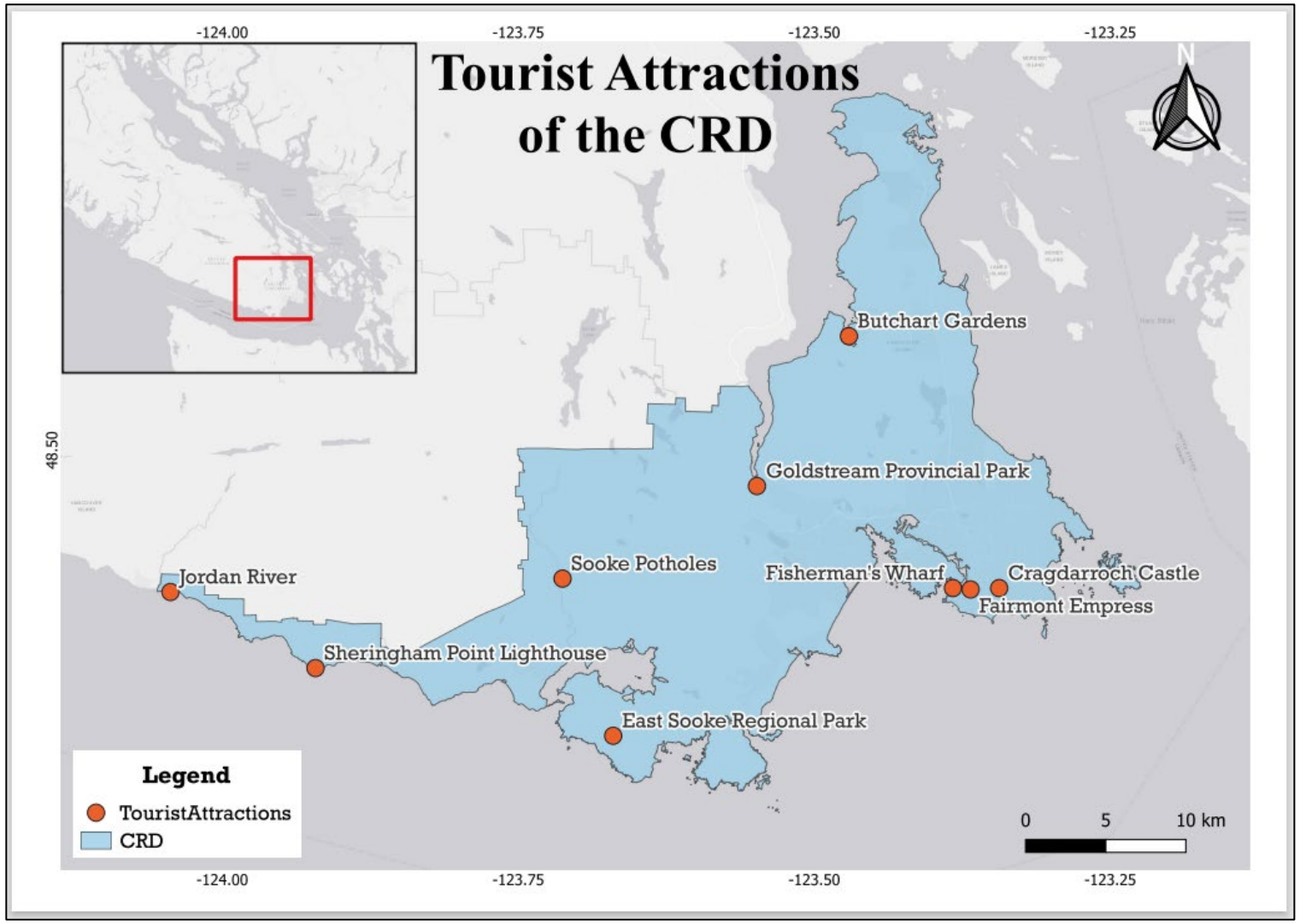
Map created on *Google Earth Pro*

***QGIS* offers basic map "decorations" on map view**:
can screenshot or export as PDF or image
But can't add…

- Legend
- Shapes
- Overview inset map



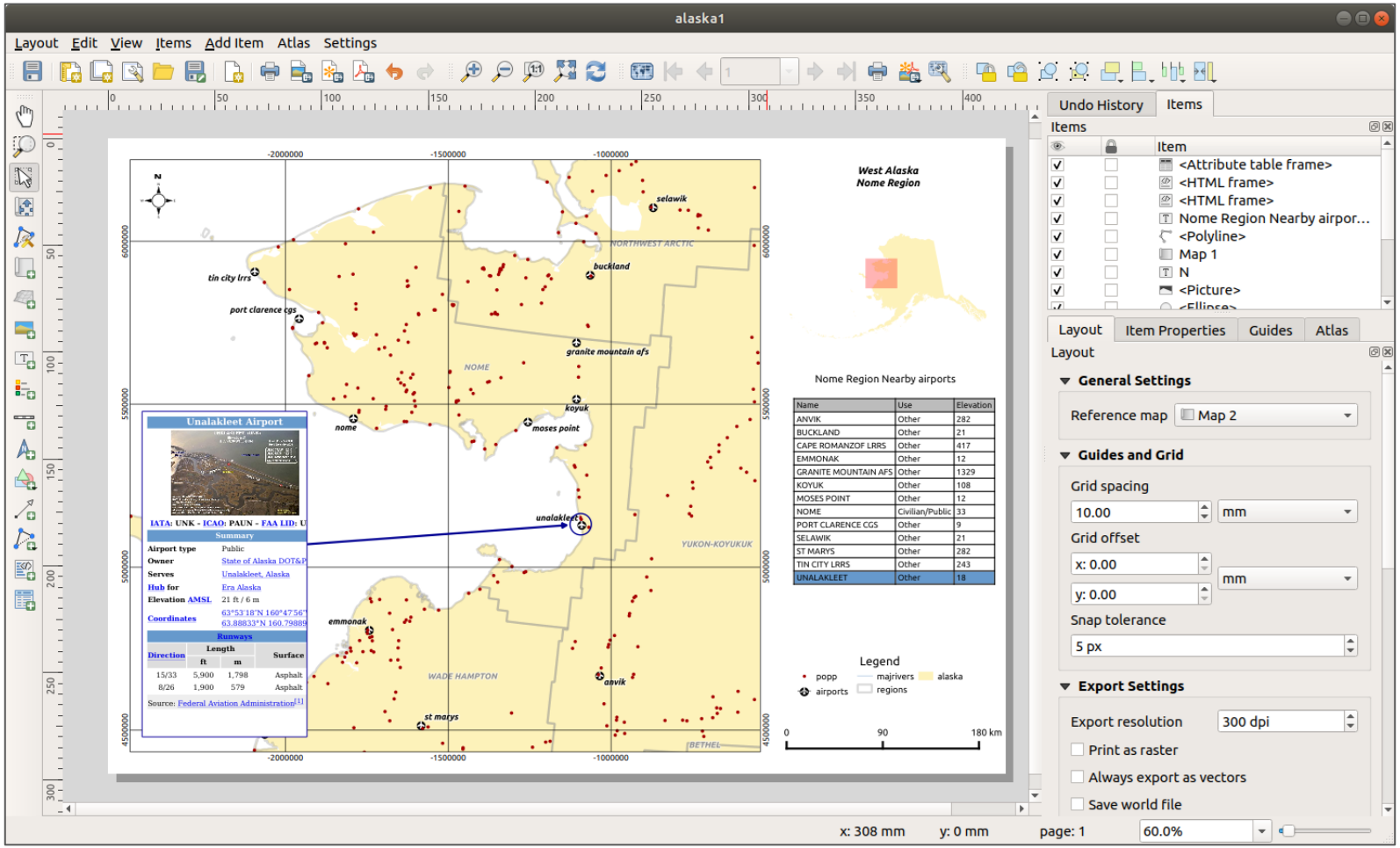
# *QGIS Print Layout* can create advanced custom maps




Map created on *QGIS* ***Print Layout***

# Exporting map on *QGIS*

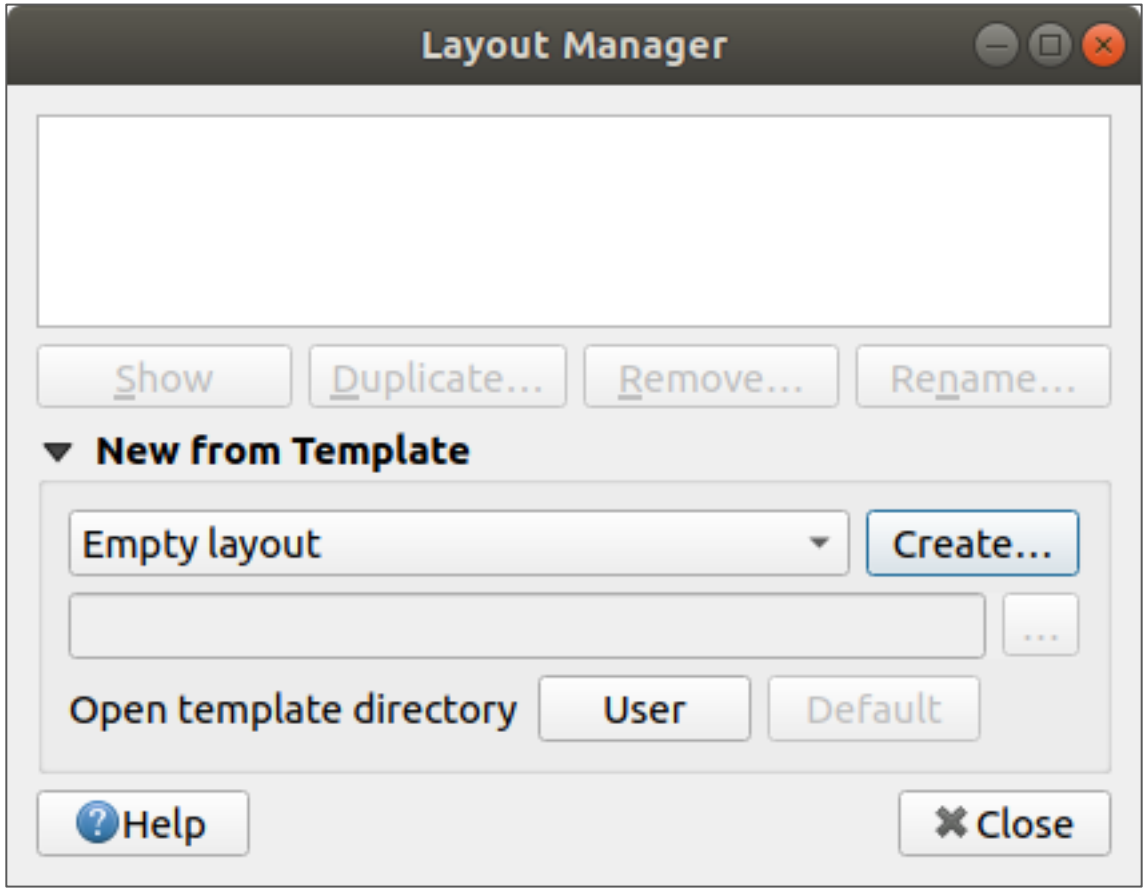
- Need to be able to print or export *QGIS* map
- GIS map file is not an image - has references to all layers, labels, colours, etc.
- Can export map file via *Print Layout*




QGIS Documentation

# Why *Print Layout*?

- Can create and export multiple maps for the same *QGIS* map
    - Organize these using *Layout Manager*
- Can create and save templates for future maps

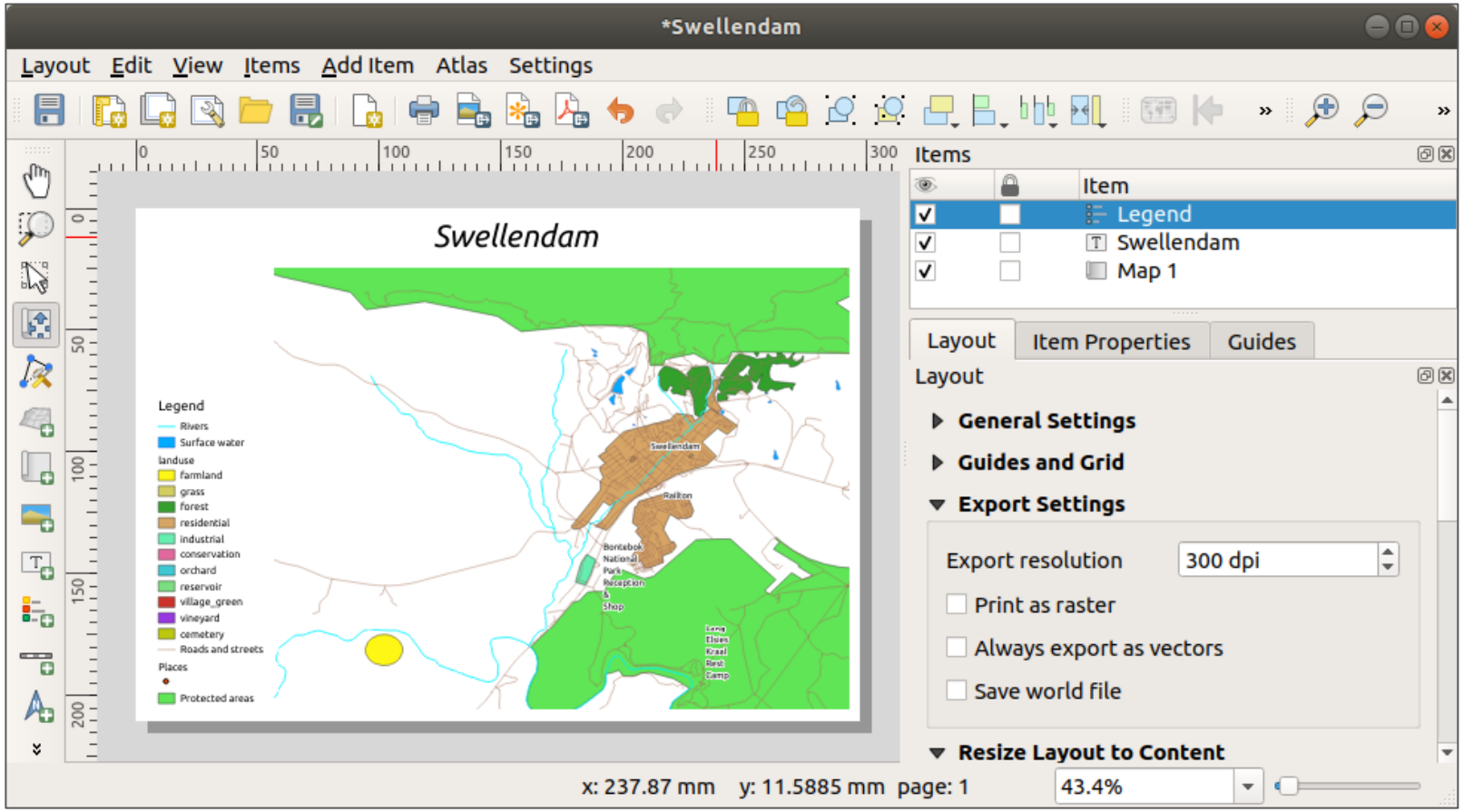



QGIS Training Manual

# Why *Print Layout*?

*Print Layout* allows:

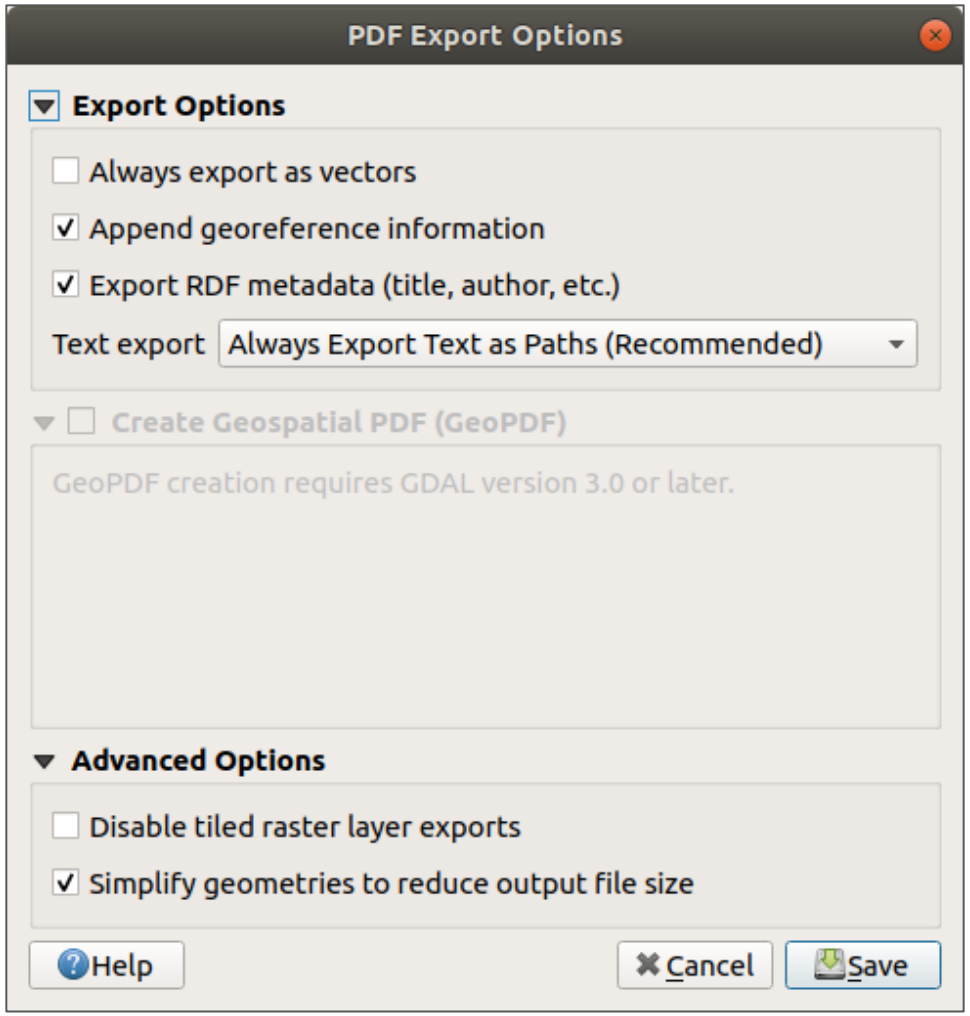
- customizable legends
- text
- images
- overview inset maps
- grids/coordinates
- shapes
- etc.




QGIS Training Manual

# Why *Print Layout*?

- Can export maps in multiple formats
  - .pdf, .svg, .jpeg, .png, etc.




QGIS Training Manual

# Understand *QGIS* interface

*your interface may look different

Menu Bar

Toolbar

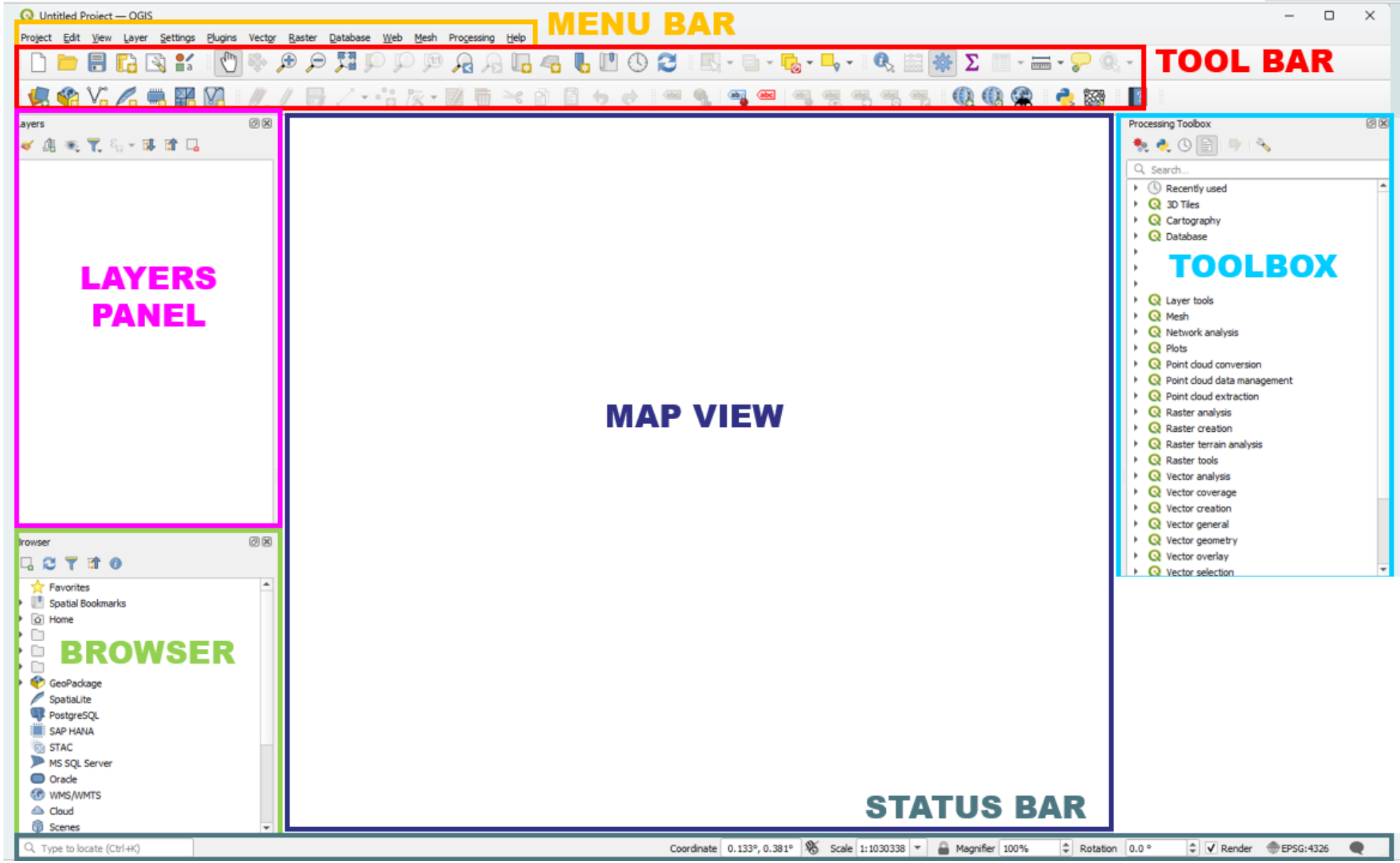
Layers Panel

Browser

Map View

Status Bar

Toolbox



# Understand *QGIS* interface

**Menu Bar:** Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh HCMGIS Processing Help

Horizontal bar at top provides access to various functions and tools

*(Project management, Edit, Plugins, Vector & Raster tools, etc.)*

**Toolbar:**

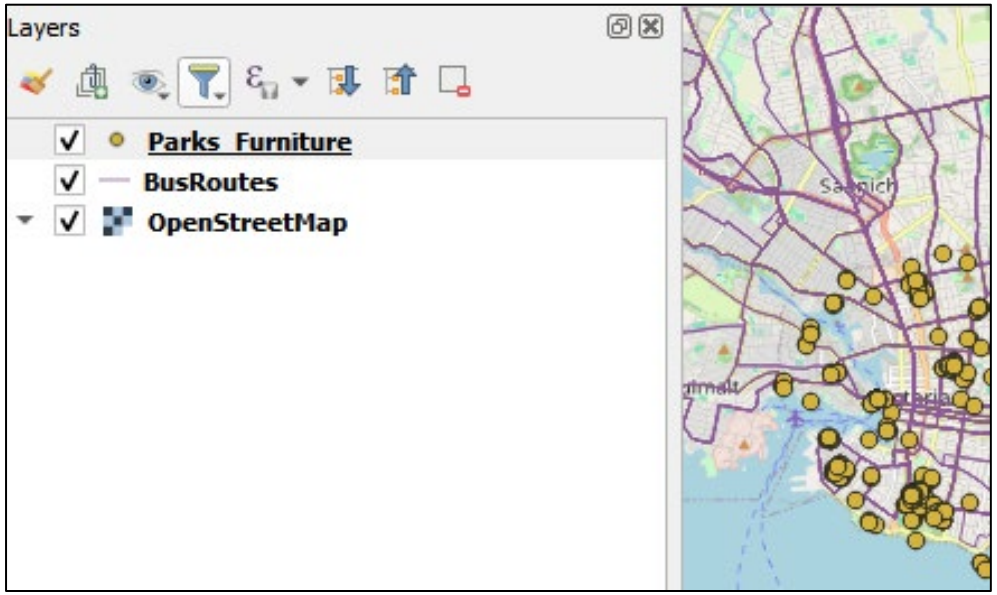
Contains icons for frequently used tools, such as Add Data, Pan, Zoom, Identify, etc.

Quick access to essential operations.

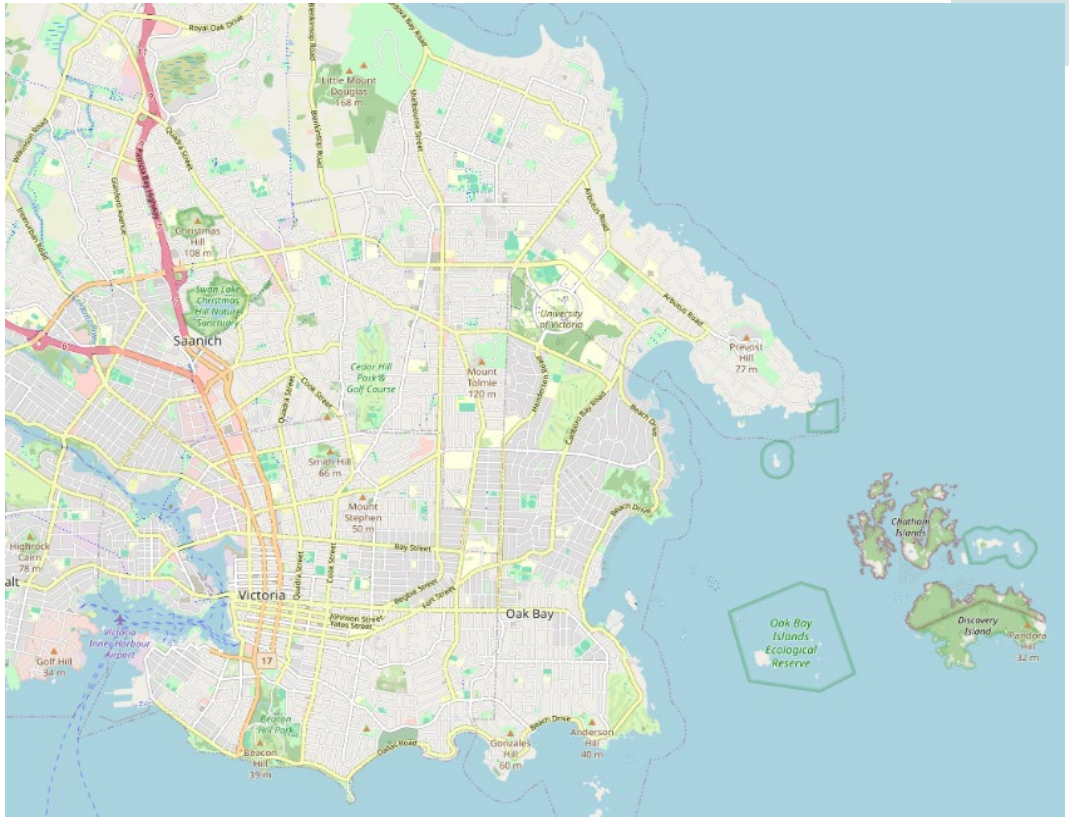
**Layers Panel:**

**Displays** all active **Layers** in the project.

Allows users to organize, manage visibility and access properties of layers



# Understand *QGIS* interface



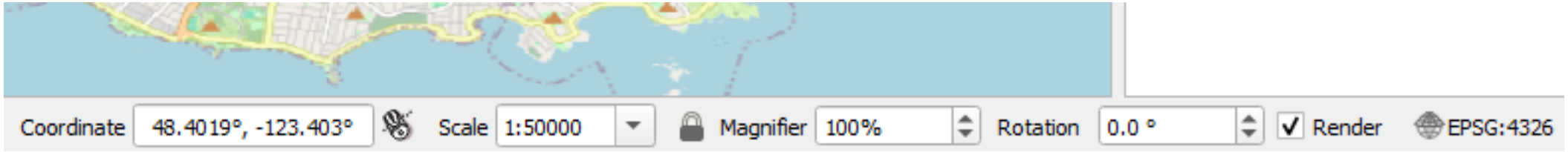

**Map View:**

The central area where spatial data is displayed.

Users can interact with the map, visualize layers and analyze spatial relationships.

**Status Bar:**

Located at the bottom, it provides information about current project: coordinate display, scale and CRS settings.



# Toolbar essentials

New project, open, and save

Pan

zoom

**“zoom full”**

Add data

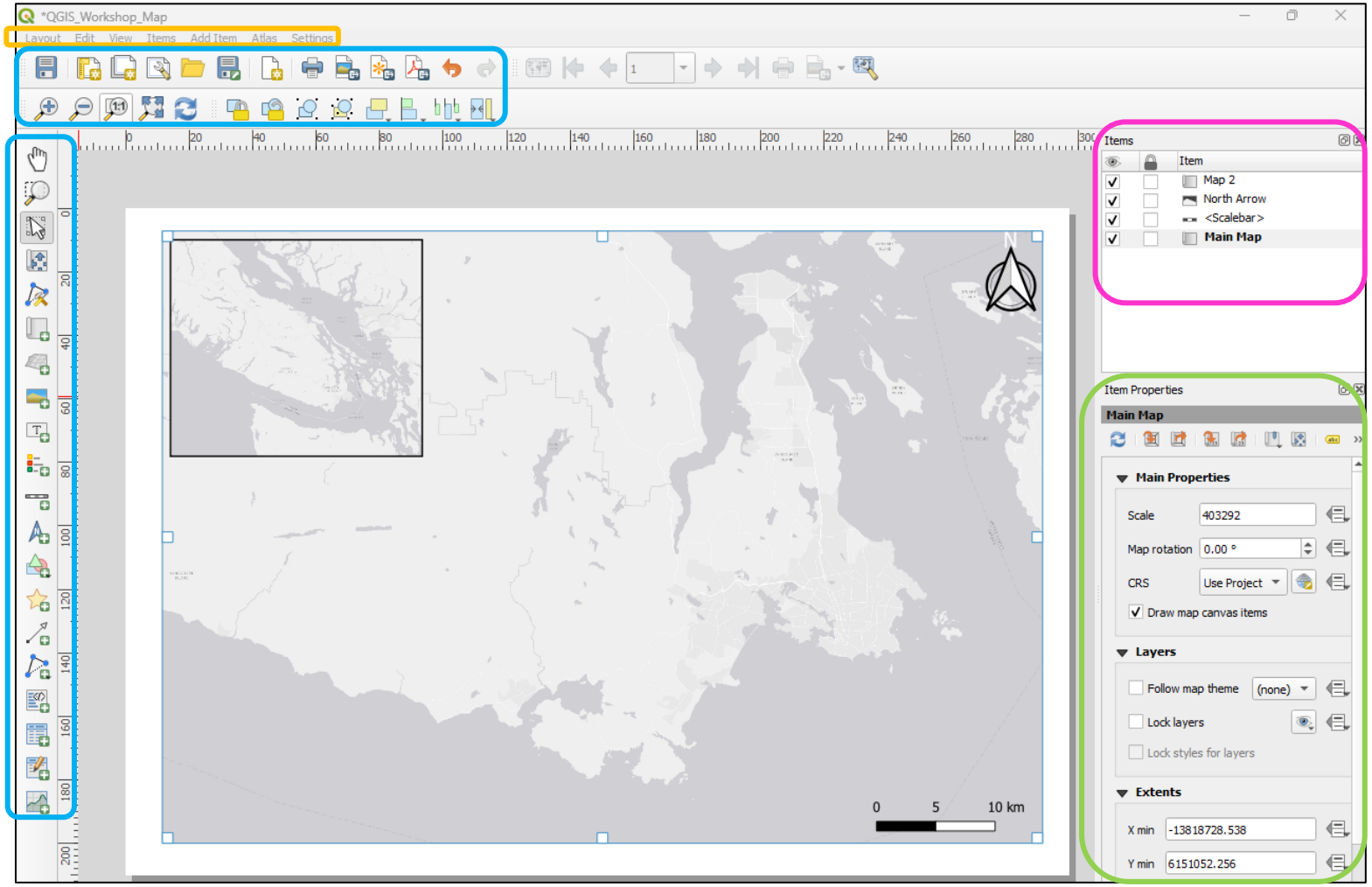
# Print Layout interface

***Menu Bar*:** Layout (save, templates, export, etc.), Add Item, Settings, etc.

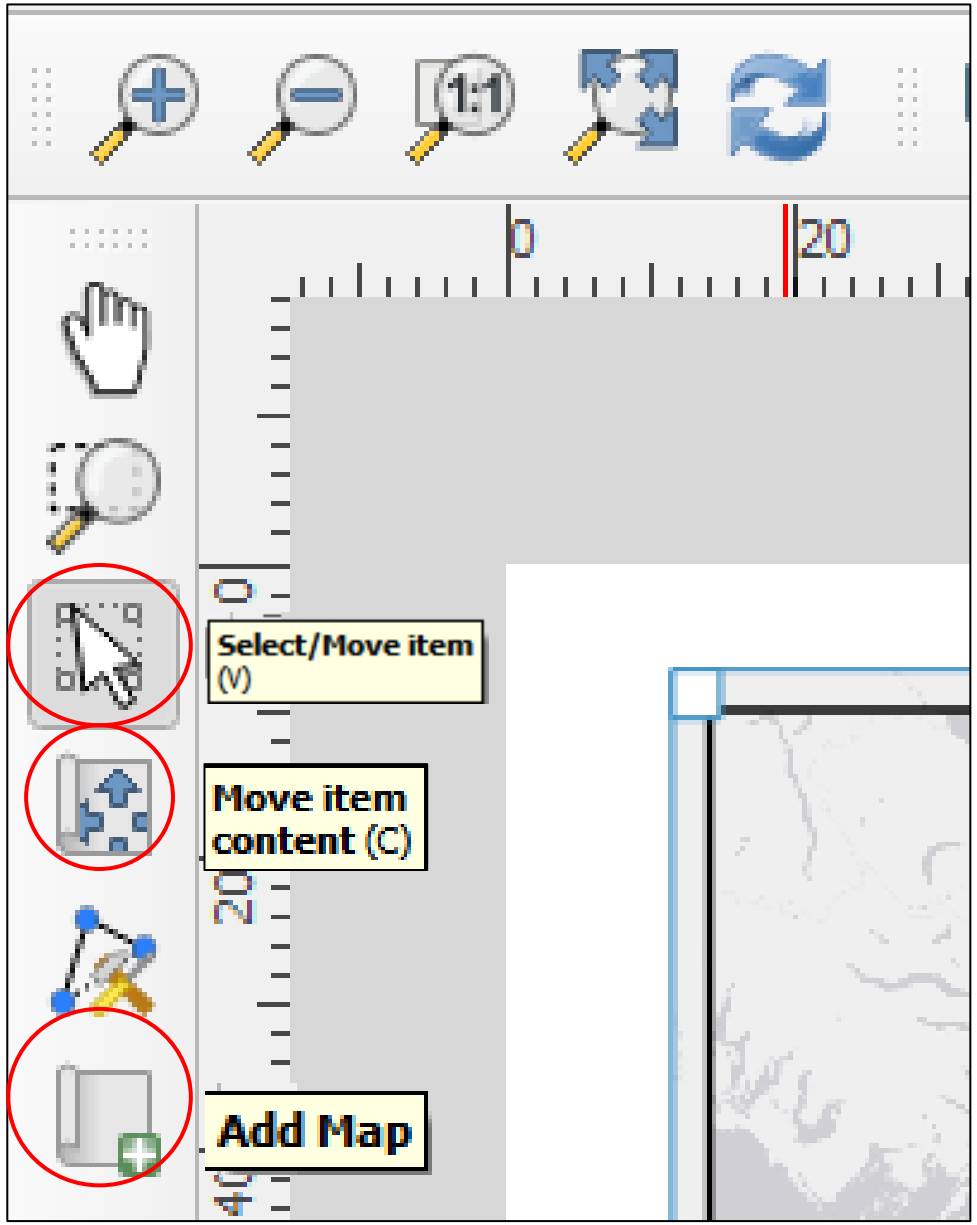
***Toolbar(s)*:** essential tools - save, open, export, zoom, add items, select/move item, etc.

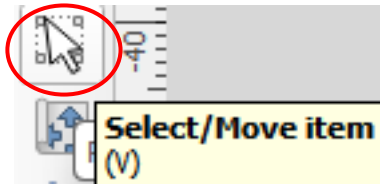
***Item Panel***: items on the map – toggle on and off, lock item, change item name, rearrange items
select item from the *Item Panel* to see its properties

***Item Properties***: properties of selected item– many options for changing scales, fonts,, borders/frames, units, etc.



# Print Layout interface: tools




*Select/Move Item:* move items in the print layout, opens the selected items' *Item Properties*

*Move Item Content:* move the map itself

Add map to print layout:

click and drag to draw map on white canvas

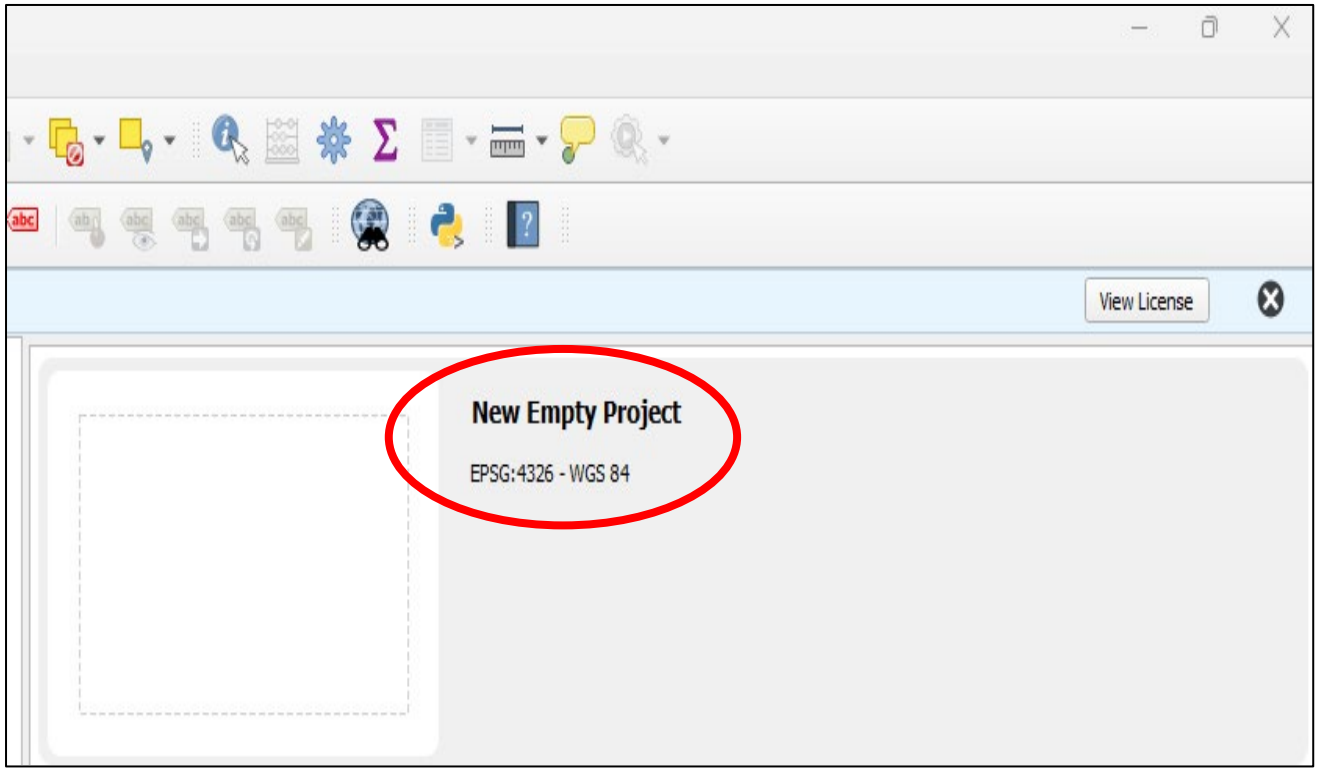
# Start *QGIS* *(if you haven't already)*

**Download workshop data**

- Extract /unzip the .zip file
- Save it where you can find it…

**Open QGIS** (your version may be different)

- Double click on *New Empty Project*




**Note**: New *QGIS* projects open with Geographic Coordinate System (GCS) **EPSG:4326**

Continue to *Workshop Exercise*

From this point onwards
**everyone works at their own pace**

There are **Check-In** slides to **ask** for help
(or **ask** sooner!)